Integrated Multiomics Reveals Alterations in Paucimannose and Complex Type *N*-Glycans in Cardiac Tissue of Patients with COVID-19

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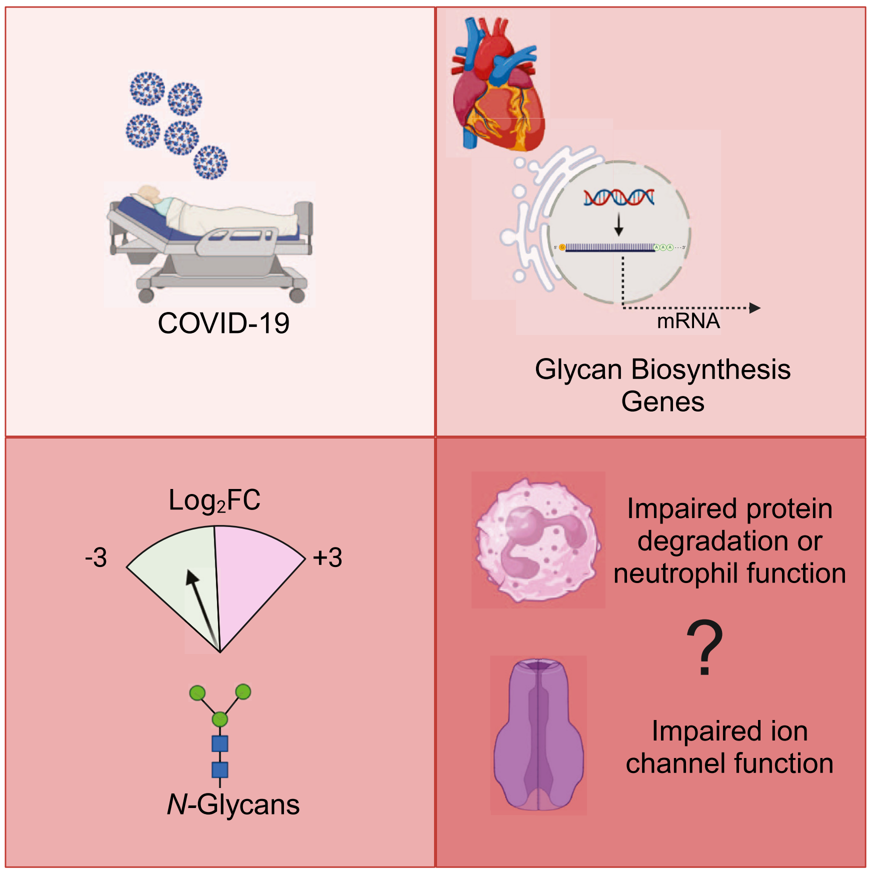
In Brief

This study identified significant expression differences in glycogenes involved in *N*-glycan biosynthesis and a reduction in high mannose and isomers of paucimannose structures in COVID-19(+) hearts. These observations suggest that SARS-CoV-2 infection primes cardiac cells to alter the glycome with an overall trend to favor synthesis and reduce degradation. As *N*-glycosylation is critical for cardiac function, this work provides a basis for understanding the molecular events leading to cardiac damage post-COVID-19, and in the future may be extended to other viral insults and organs.

Graphical Abstract



Highlights

- Mass spectrometry and RNAseq reveal alterations in the glycome of COVID-19(+) hearts.
- COVID-19(+) hearts have altered expression of glycogenes for *N*-glycan biosynthesis.
- High mannose and paucimannose structures are decreased in COVID-19(+) hearts.

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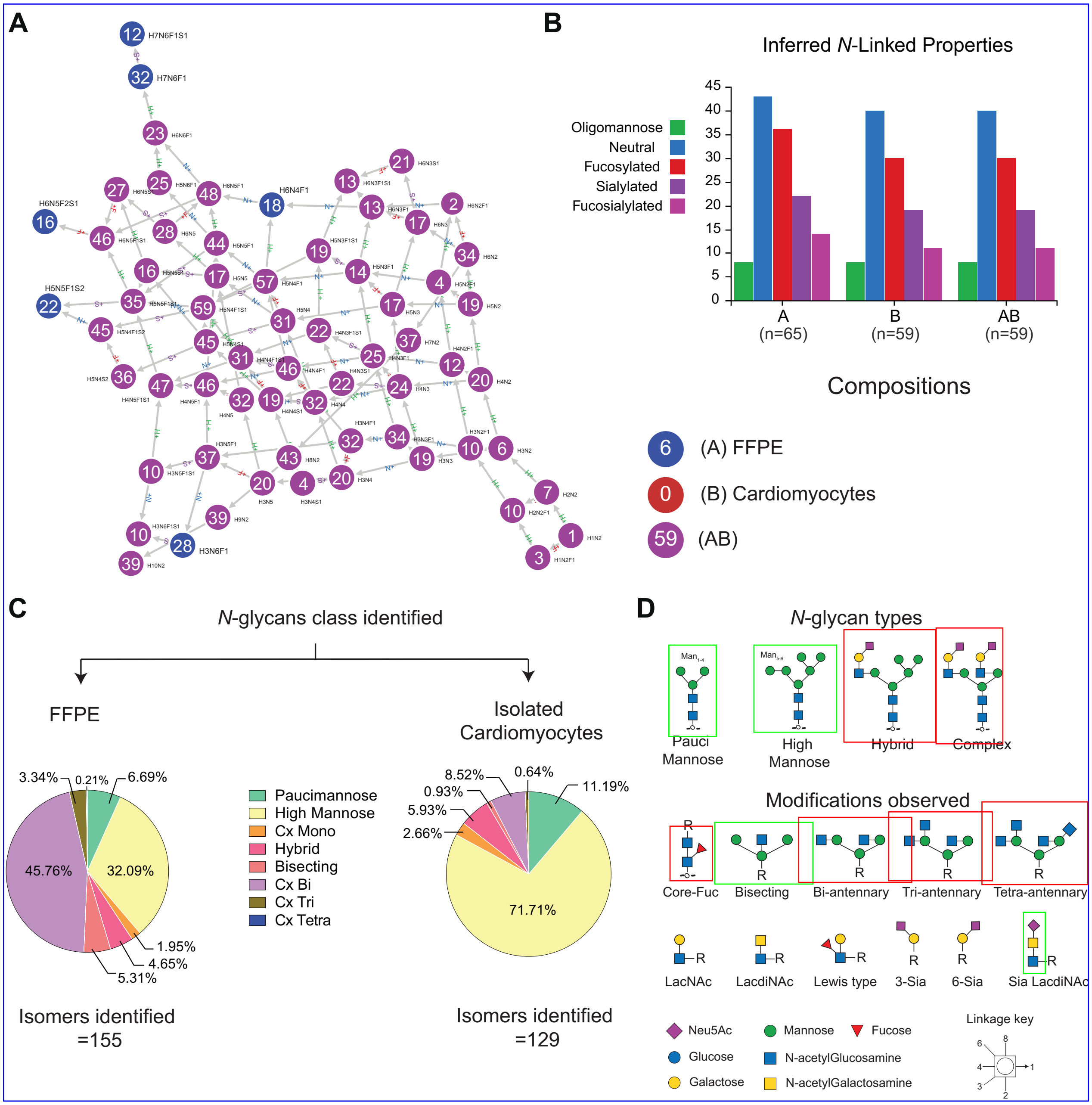

FIG. 3. **Qualitative *N*-glycan profile of cardiac tissue and isolated cardiomyocytes.** *A*, graph of relatedness of observed *N*-glycan compositions identified in FFPE cardiac tissue slices and isolated cardiomyocytes. Each node represents a unique glycan composition, and two adjacent nodes differ by one monosaccharide residue. *Blue* nodes are observed only in FFPE tissue, and *purple* is observed in both FFPE tissue and isolated cardiomyocytes. *B*, distribution of inferred *N*-glycan composition properties. Graphs in *A* and *B* were generated using Glycompozitor (31). *C*, Pie charts summarizing the percentage distribution of each *N*-glycan class calculated from the relative abundance values. *D*, graphical representation of glycan types identified, and modifications observed at the core and terminal branches.

(~12%) *N*-glycans were significantly different between COVID-19(–) and COVID-19(+) groups (Fig. 4*B*). Isomers of paucimannose, high mannose, and hybrid-type glycans were significantly reduced and LacDiNAc-type structures in the complex glycan class were elevated in the COVID-19(+) group.

Unlike FFPE cardiac tissue, *N*-glycan compositions in isolated cardiomyocytes had mixed patterns with few paucimannose, high mannose, and complex structures differing in abundance (supplemental Tables S7 and S8). Specifically, *N*-glycans with compositions corresponding to paucimannoses (H1N2, H4N2, H4N2F1), high mannoses (H5N2, H5N2F1,

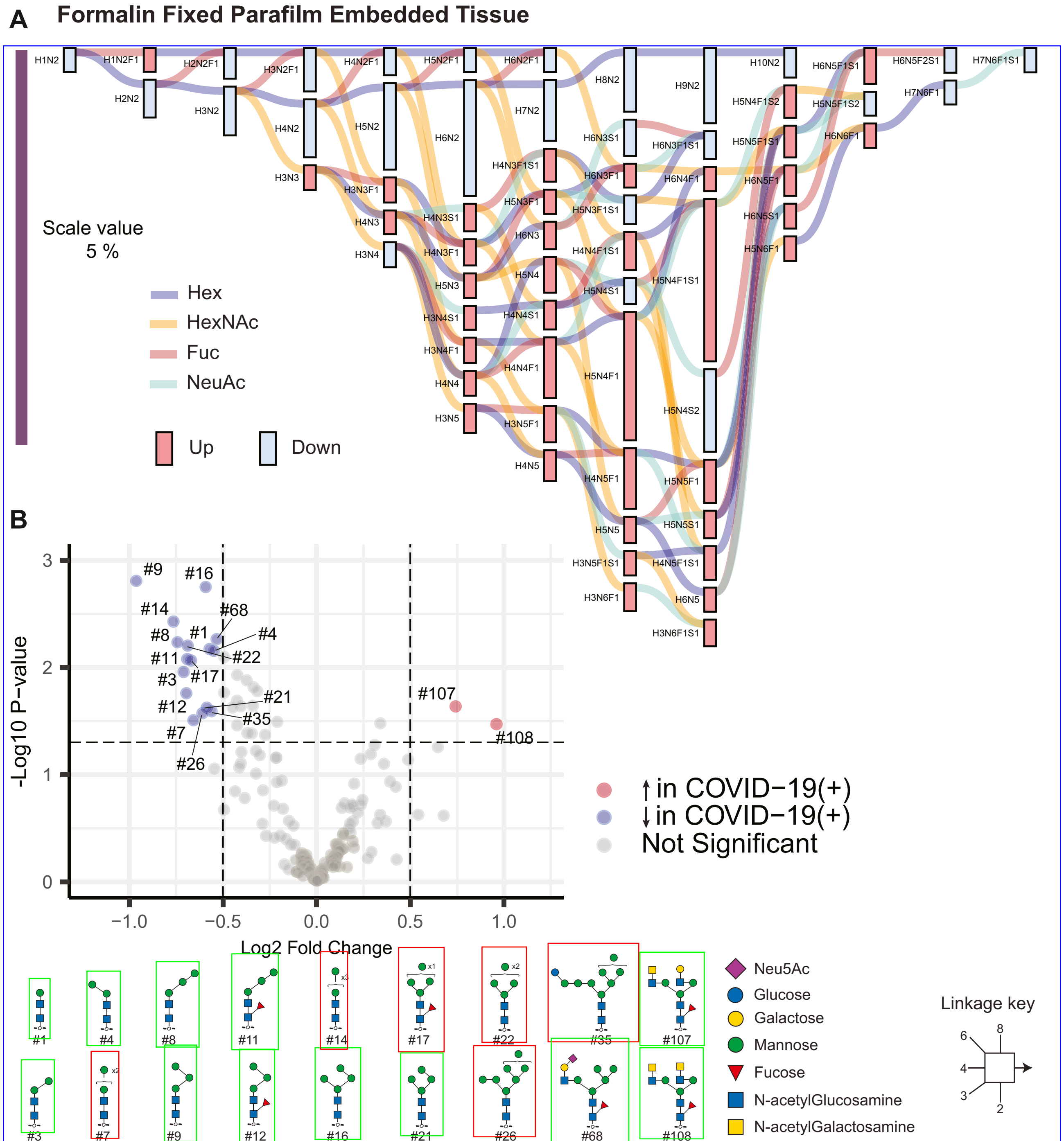

Fig. 4. **Quantitative differences in the *N*-glycome of cardiac tissues (FFPE) from COVID-19(+) and COVID-19(−) patients.** *A*, compositional analysis of *N*-glycans in cardiac tissue. Graphical results display output from differential expression analysis using GlynSight. Hex (H), HexNAc (N), Fuc (F), and Sia (S) are monosaccharide moieties that form glycan structures. *Red* and *blue* bars depict differences in relative abundance. *Colored lines* indicate the addition of monosaccharide moieties to a given composition. A downward trend (*blue bar*) in H_nN_2, corresponding to paucimannose, and oligo mannose indicates reduced levels, and an upward trend (*red bars*) corresponding to H_nN_nFS for complex and hybrid structures, indicates an overall increase in abundance. *B*, Volcano plot displaying *N*-glycan structures that are most significantly different between cardiac tissues from each patient group.

are a different patient cohort than used for glycomics; therefore, a direct correlation between glycan structures found and glycosyltransferase/hydrolase expression and their distribution in individual cell types could not be established. Second, despite differences in sample size and nuclei number, some of the differences in glycogene expression observed in the left ventricle by Delorey *et al.* (20), when compared to the right ventricle reported by Brener *et al.* (19), suggest a region-specific glycogene expression as one study used the left ventricle and the other used the right ventricle tissue. However, differences could also be due to different donor pools, inherent differences in cellular distribution, and/or the extent of infection across anatomical regions. As data comparing multiple regions within donors are not available, future analysis of age, sex, and anatomical site-matched samples of the human heart are necessary to determine if cardiac glycogene expression differences are regional. Finally, underlying comorbidities within the cohort used for our study add to the sample heterogeneity and could only be mitigated *via* statistical analysis. Given the practical challenges to biobanking during the pandemic, inclusion and exclusion criteria were not established before collection and the cohort sizes were modest, although consistent with many previously published studies of COVID-19 (19, 20).

In summary, we expanded on our prior *N*-glycome dataset for the human heart, which we expect will inform future studies designed to understand cellular and molecular changes occurring in cardiac disease. Using this library, we observed changes in *N*-glycan structure abundance that occur in the human heart post-COVID-19. Major changes in paucimannose and complex structures were observed in COVID(+) cardiac tissues. Our observations suggest that SARS-CoV-2 infection primes cardiac cells to alter the glycome at all levels, namely metabolism, transport of nucleotide sugar, and glycosyltransferases with an overall trend to favor synthesis and reduce degradation. These changes can be attributed to the viral life cycle (*e.g.*, hijacking host machinery) or cellular transformations occurring during infection. As *N*-glycosylation is critical for maintaining electrical conductivity, cell adhesion, and protein quality control in the heart, this work provides a basis for understanding the molecular events leading to cardiac damage post-COVID-19, and in the future may be extended to other viral insults and organs.

DATA AVAILABILITY

Raw files from MS has been deposited in GlycoPOST (52). (https://glycopost.glycosmos.org/preview/1409704126670147788a822. Pin=7380).

Supplemental data—This article contains supplemental data (53).

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Conflict of interest—R. L. G. is on the advisory board of ProtiFi, LLC, and receives no compensation of any kind.

Abbreviations—The abbreviations used are: AAL, aleuria aurantia lectin; COVID-19, Coronavirus infectious disease of 2019; ECL, erythrina cristagalli lectin; FDR, false discovery rate; FFPE, formalin fixed paraffin embedded; Fuc (F), fucose; H and E, hematoxylin and eosin; Hex (H), hexose; hexnac (N), N-acetyl hexosamine; MAL, maackia amurensis lectin; mnesi, multinozzle electrospray ionization; MS, mass spectrometry; PGC, porous graphitized column; pngase F, peptide N-glycansase F; PNA, peanut agglutinin; RCA, ricinus communis agglutinin; SARS-cov-2, severe acute respiratory syndrome coronavirus 2; Sia (S), sialic acid; snRNA-seq, single nuclei RNA sequencing; SNA, sambucus nigra agglutinin; VVA, vicia villosa lectin.

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